

Could you please elaborate a bit on the 3D-IC?

Essentially, in a 3D-IC, we stack different dies in a single package. The silicon dies are connected through what we call 'silicon interconnects,' which have tens of thousands of connections between each die. So, it essentially becomes a package as it is stacked side-by-side.

The variety of things that we can put on these chips is immense—we can put in algorithms, software, microprocessors, inputs/outputs, protocols, logics and so on. So we have also integrated, along with FPGA, a transceiver die. This is the die that helps us to go to the 28Gbps bandwidth—the highest bandwidth at the moment among FPGAs.

3D-ICs were shipped along with programmable SoCs in Q4 of 2011. Having this integration allowed us to increase the gate capacity. We were able to get two million logic cells onto a single device. Just to put it in perspective, that's about 20 million ASIC kits. Our nearest competitor can only do half of that amount. The only reason for that is our 3D-IC capability.

What challenges did you face while working on 3D-IC technology?

The 3D-IC has been a different ball-game altogether. Our research group inside Xilinx has been working on this product for the last five years, so we have a very clear idea about what we are doing. It is not enough to just come up with a chip, we need to make it manufacturable too. In order to do this, our team worked with universities, fab partners and packaging vendors to essentially come up with an ecosystem.

Another thing to note is that if we are the only company to do this, then it is not good for us. So we have worked with semiconductor companies as part of the semi group that we have and they have also adopted this technology.

What were the key issues that were solved by utilising the new lower process node?

The key theme when we developed this was power. Everybody thinks that FPGAs are power-hungry, but we have been able to solve that problem significantly. By going from 40 nm to 28 nm, we have reduced the power consumption to almost half. The all-programmable FPGAs are clearly a generation ahead in terms of memory bandwidth, transceiver performance, DSP performance and integration capability.

What worked in your favour in designing this portfolio?

We partnered with a fab (TSMC) and chose a process called 28nm HPL (high-performance low-power). Usually, a lot of resources go into choosing the right process. Our R&D team took a bet on HPL because they thought it was the best process. The other options were HP (high-performance) or LP (low-power). This was a process that we co-developed with our fab partner.

With the coming of 28 nm, what major changes were required in your software tools?

One of the reasons why FPGA companies are so successful is their software and, on the other hand, it is the same reasons why startup FPGA companies have not been so successful.

At Xilinx, ISE used to be our traditional design suite. Now, at 28 nm, the complexity of our devices has exponentially grown, and so our tools have no longer been able to keep pace with the same. For example, implementing a design takes hours. If it is a large design, we have had cases where the customer had been implementing this design for eight hours, compiling and routing, etc.

So Xilinx decided to get a complete new tool suite that can cope with all the issues including run-time and quality of results.

Now that Xilinx is in 20 nm domain, tell us what are the key differences between 28 nm and 20 nm?

From an FPGA standpoint, transceivers are now system-optimised. Thus transceivers are taking care of the entire system. We have more than a hundred transceivers per chip running at 33 Gbps. We have DSP blocks and block RAMs inside, so those will be faster now. So far we have been supporting DDR1, DDR2 and DDR3 memories. But now DDR4s are coming up, so we will support them as well. This allows us to give twice the memory. In terms of routing architecture, this will allow more than 90 per cent routings. This is enabled not just by the routing architectures but also by the tools.

Traditionally, most FPGAs struggle with 70-80 per cent routing but we are able to utilise 90 per cent of the routing capabilities. We have optimised power utilisation so that it uses only half of what it has been using, besides giving some more granular optimisation for the tool to optimise your power. We have enhanced the mixed-signal capability two-fold.

We are now offering heterogeneous multi-core SoCs (SoCs with asymmetric cores). Performance is optimised as we have multi-core, memory and fabrics. So we are going to have more optimisation across the family so that the transactions between the processor and the FPGA are much more efficient. Power optimisation is a key factor in this segment too. There is a lot of integration done so that it reduces the bill-of-materials cost for the customer.

And in 3D-ICs, we have always had FPGAs and transceivers, but now we also have wide memory. What we are going to offer is memory, FPGA and transceivers. We will be offering industry-standard interface between dies so that if there is another ASIC from a customer that needs to be integrated, we can offer that.

We are offering cutting-edge functionality with future transceiver protocols being supported up to 56 Gbps. The density will be enhanced by 50 per cent to three million logic cells, which is equivalent to about 30-40 million ASICs. ●

(The complete interview can be read at electronicsforu.com)